

Findings & Priority Segments

Business Analyst Case Study | Dataset: Olist E-Commerce (seller marketplace)

05 · FINDINGS & PRIORITY SEGMENTS

Summary of Findings

Finding	Status	Evidence
Seller tenure predicts cancellation risk	Confirmed	1.08% (new) vs 0.40% (established); $p < 0.001$
Tenure effect holds within categories	Confirmed	Automotive: 2.51% (new) vs 0.51% (established)
Product category alone is a weak driver	Weak / inconclusive	High-rate categories are low-volume outliers
Delivery delay predicts cancellation	Ruled out	Data structurally excludes cancelled orders from this cut

Category Breakdown (Top Segments by Rate, $n > 50$ orders)

Category	Total Orders	Problem Orders	Rate
dvds_blu_ray*	59	2	3.39%
construcao_ferramentas_seguranca	167	5	2.99%
construcao_ferramentas_jardim	194	4	2.06%
instrumentos_musicais	628	11	1.75%
telefonias_fixa	217	3	1.38%
livros_interesse_geral	512	7	1.37%
eletrodomesticos_2	234	3	1.28%

*Flagged as a low-confidence signal: on only 59 orders, this rate is highly sensitive to a single data point and should not be treated as a reliable pattern (see 02_Hypotheses_and_Examples.pdf for the worked example on why).

Seller Tenure Breakdown

Seller Tenure Bucket	Total Orders	Problem Orders	Cancellation Rate
0-30 days (new)	12,253	132	1.08%
30-90 days	16,451	85	0.52%
90+ days (established)	83,946	332	0.40%

New sellers show a rate 2.7x higher than established sellers, statistically validated at $p < 0.001$ (full calculation in 04_Statistical_Validation.pdf).

Priority Matrix: Category × Seller Tenure

Combining category and seller tenure surfaces which specific segments deserve first attention, ranked by an impact score (order volume × cancellation rate) — this balances "how bad" a segment is against "how many orders" it actually affects.

Category	Seller Type	Order Volume	Cancel Rate	Impact Score
esporte_lazer	established	6,896	0.62%	4,300
informatica_acessorios	established	5,934	0.72%	4,300
utilidades_domesticas	established	5,104	0.84%	4,300
beleza_saude	established	7,731	0.39%	3,000
moveis_decoracao	established	5,949	0.45%	2,700
brinquedos	established	3,500	0.71%	2,500
automotivo	new_seller	557	2.51%	1,400
cama_mesa_banho	established	8,788	0.18%	1,600

Note: the top three rows by raw impact score are large, high-volume, established-seller categories with average performance — not action targets. The standout signal is automotivo / new_seller: despite lower volume, its rate is nearly 5x the established-seller rate within the same category, making it the sharpest, most specific target for intervention.